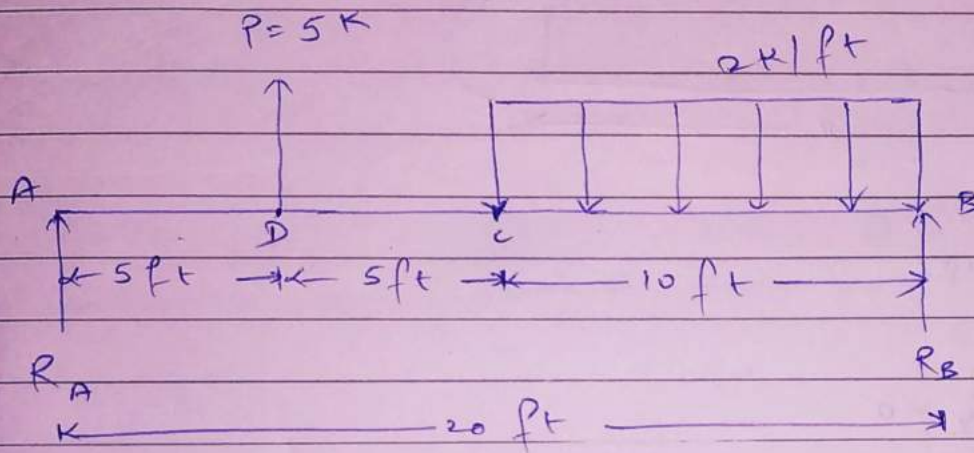


Solution Que: ① :-

FBD of the Beam:-



Considering equilibrium & drawing FBD, taking moment about pts (A) equal to zero.

$$\sum M_A = 0$$

$$(5 \times 5) + (R_B \times 20) - (2 \times 10 \times 15) = 0$$

$$R_B = 13.75 \text{ Kips}$$

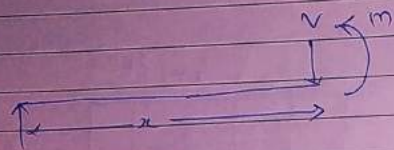
$$\sum F_y = 0$$

$$R_A + 5 - 2 \times 10 + 13.75 = 0$$

$$R_A = 1.25 \text{ Kips}$$

(2)

⇒ Taking a section (xx) b/w nD at a distance (x) from (A).



$$P_A = 1.25 \text{ Kips}$$

$$\sum F_y = 0$$

$$-V + 1.25 = 0$$

$$V = 1.25 \text{ Kips} \rightarrow \text{Shear force acting on the section}$$

$$\sum M_x = 0$$

$$M - 1.25 \times (x) = 0$$

$$M = 1.25x \rightarrow \text{Bending moment at the section}$$

at $x=0$

$$V_A = 1.25 \text{ Kips} \quad M_A = 0 \rightarrow \text{Hinge Support}$$

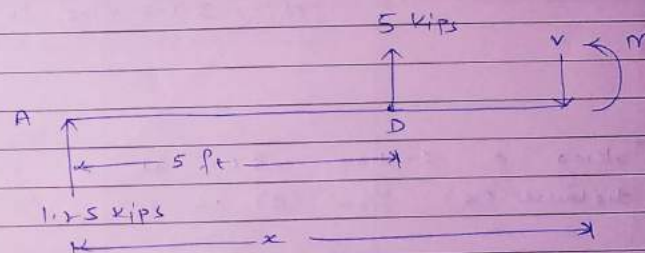
(3)

at $x = 5 \text{ ft}$

$$V_D = 1.25 \text{ Kips} \quad ; \quad M_D = 1.25 \times 5$$

$$M_D = 6.25 \text{ Kips. ft}$$

⇒ Taking a section b/w DC at a distance x from A :-



$$\sum F_y = 0$$

$$-V + 1.25 + 5 = 0$$

$$V = 6.25 \text{ Kips}$$

$$\sum M_x = 0$$

$$M - 5(x-5) - 1.25 \times x = 0$$

$$M = 5x - 25 + 1.25x$$

$$M = 6.25x - 25$$

(4)

at $x = 5 \text{ ft}$:-

$$V_D = 6.25 \text{ Kips} \quad ; \quad M_D = 6.25 \times 5 = 25$$

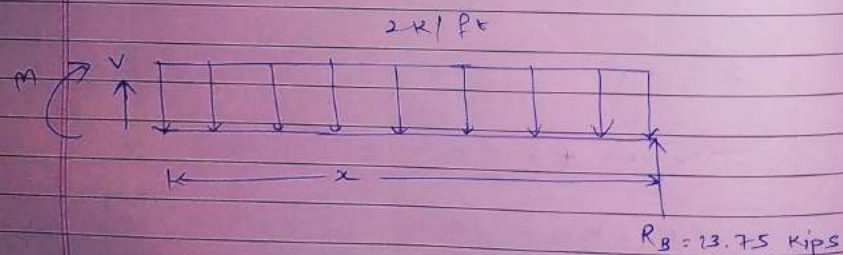
$$M_D = 6.25 \text{ Kips-ft}$$

at $x = 10 \text{ ft}$:-

$$V_C = 6.25 \text{ Kips} \quad ; \quad M_C = 6.25 \times 10 = 25$$

$$M_C = 37.5 \text{ Kips-ft}$$

$\Rightarrow$ Taking a section BC at a distance (x) from (B) :-



$$\sum F_y = 0$$

$$V + 13.75 - 2x = 0$$

$$V = 2x - 13.75$$

(5)

$$\sum M_A = 0$$

$$-M + 13.75 \times x - 2 \times (x) \times \left(\frac{x}{2}\right) = 0$$

$$M = -x^2 + 13.75x \quad \text{--- (1)}$$

at $x = 0$:-

$$V_B = -13.75 \text{ Kips} \quad ; \quad M_B = 0 \rightarrow \text{hinged support}$$

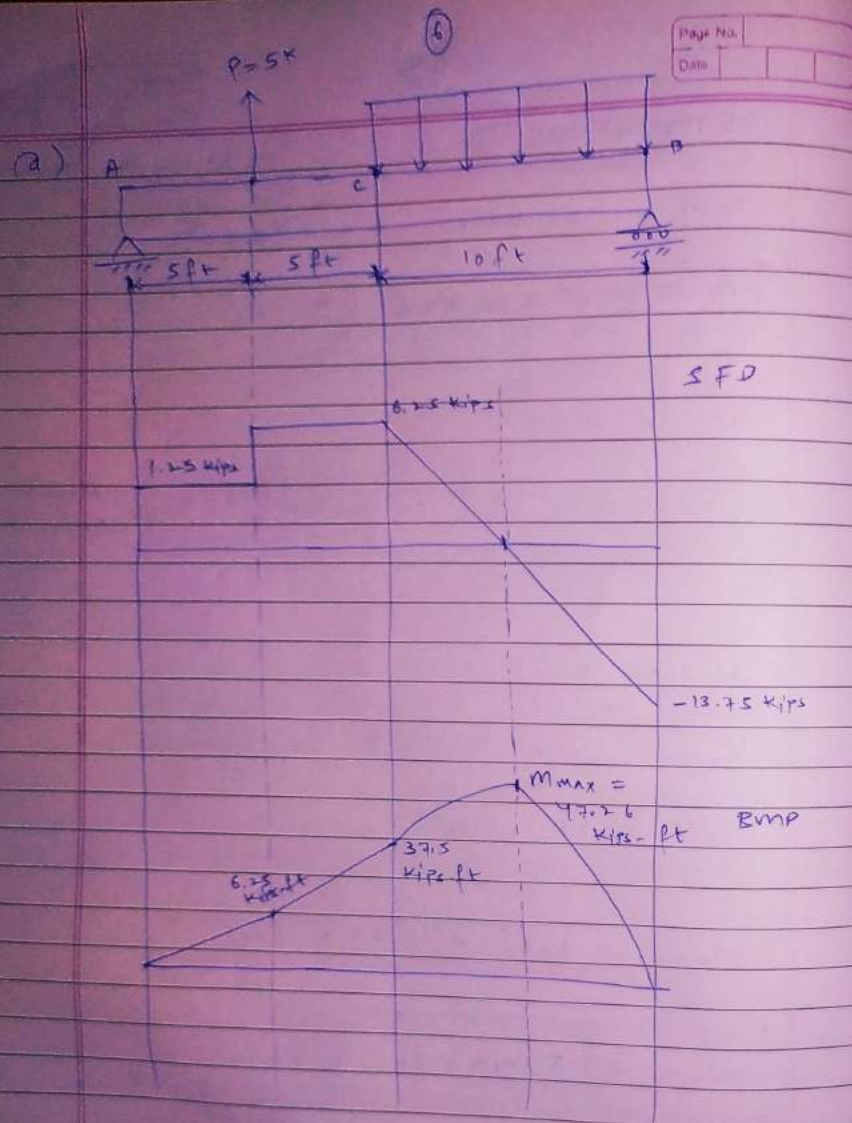
at $x = 10 \text{ ft}$:-

$$V_C = 2 \times 10 - 13.75$$

$$V_C = 6.25 \text{ Kips}$$

$$M_C = -(10)^2 + 13.75 \times (10)$$

$$M_C = 37.5 \text{ Kips-ft}$$



(b) (7)

Page No.
 Date

As the shear force reaches to zero b/w B & C. So, the Bending moment is maxⁿ at that point. To find the position of that point put $v=0$ in shear force eqⁿ for the sectⁿ b/w B & C.

$$2x - 13.75 = 0$$

$$x = 6.875 ft \text{ from B}$$

$\therefore$ Put (x) in ① :-

$$M_{max} = - (6.875)^2 + 13.75 \times (6.875)$$

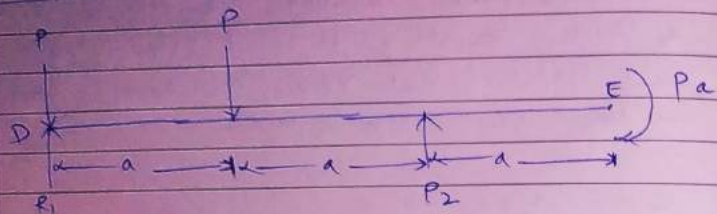
$$M_{max} = 47.26 Kips-ft$$

$$V_{max} = 13.75 Kips$$

①

Page No.	
Date	

Solution Ques ②:- (1) FBD of upper beam & let it be (DE)



Let R_1 & R_2 be the two support reactions:-

Considering equilibrium:-

$$\sum F_y = 0$$

$$R_1 + R_2 = 2P$$

$$\sum M_D = 0$$

$$-Pa - Pa + R_2 \times (2a) = 0$$

$$\boxed{R_2 = P}$$

$$R_1 = 2P - P$$

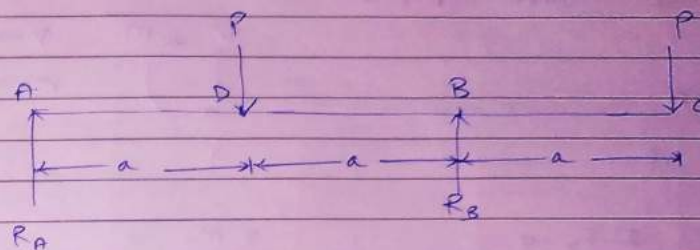
$$\boxed{R_1 = P}$$

②

Page No.	
Date	

②

FBD of Beam ABC:-



Let R_A & R_B be reactions at A & B.

$$\sum F_y = 0$$

$$R_A + R_B - 2P = 0$$

$$\boxed{R_A + R_B = 2P}$$

$$\sum M_A = 0$$

$$R_B \times (2a) - P \times (a) - P \times (3a) = 0$$

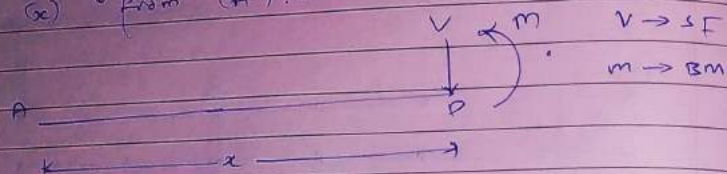
$$R_B = \frac{4Pa}{2a}$$

$$\boxed{R_B = 2P}$$

$$\boxed{R_A = 0}$$

(10)

⇒ Taking a section (AD) at a distance (x) from (A) :-



$$\sum F_y = 0$$

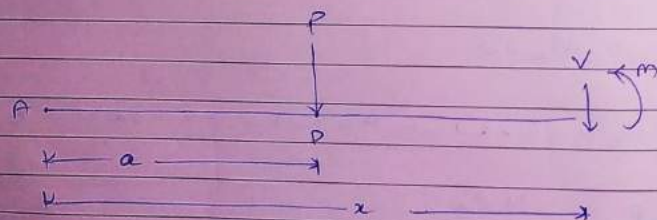
$$\sum M_x = 0$$

$$V = 0$$

$$M = 0$$

$$V_{AD} = 0 \quad \& \quad M_{AD} = 0$$

⇒ Taking a section b/w DB at a distance (x) from (A) :-



$$\sum F_y = 0$$

$$\sum M_x = 0$$

$$-V - P = 0$$

$$M + P(x-a) = 0$$

$$V = -P$$

$$M = -P(x-a)$$

(11)

at $x = a$:-

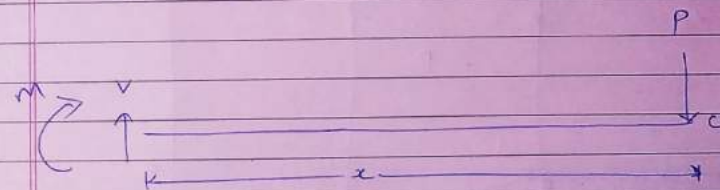
$$V_D = -P \quad ; \quad M_D = 0$$

at $x = 2a$:-

$$V_B = -P$$

$$M_B = -Pa$$

⇒ Taking a section b/w BC at x from (C) :-



$$\sum F_y = 0$$

$$\sum M_x = 0$$

$$V - P = 0$$

$$-M - P(x) = 0$$

$$V = P$$

$$M = -Px$$

at $x = 0$:-

$$V_C = P$$

$$M_C = 0$$

at $x = a$:-

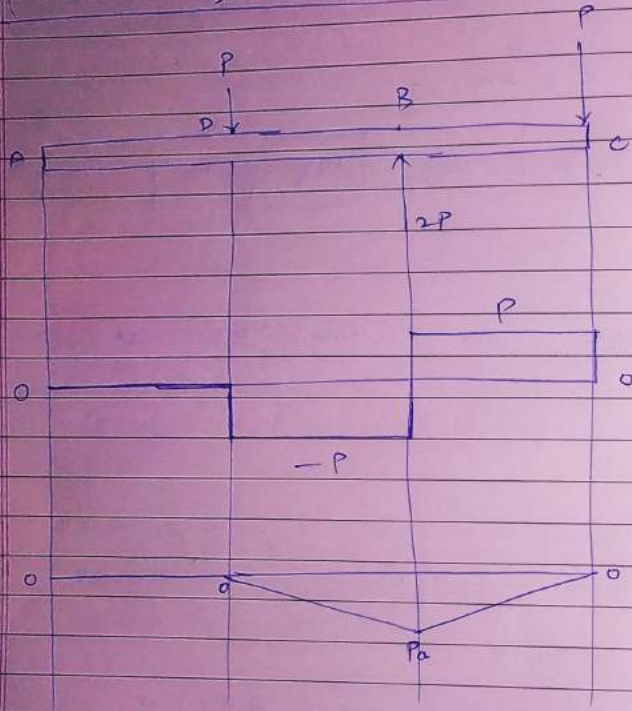
(12)

Page No.

Date

$$V_B = P \quad ; \quad M_B = -Pa$$

(SFD)



(BMD)

$$V_{\max} = P$$

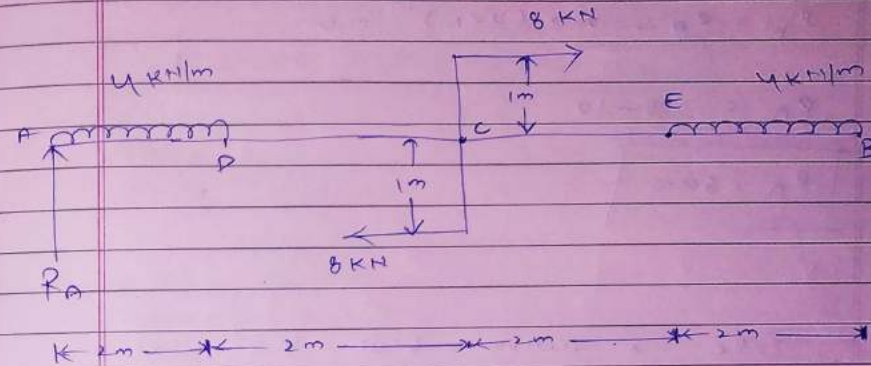
$$M_{\max} = Pa$$

(13)

Page No.

Date

Solution Que (3) :-



Developed couple = $8 (2) \text{ kN-m}$

$\therefore$; $M_C = 16 \text{ kN-m}$ can be replaced at (c). on the Beam (AB).

considering equilibrium :-

$$\sum M_A = 0$$

$$(R_B \times 8) - \left(4 \times 2 \times \frac{2}{2} \right) - 16 - \left\{ 4 \times 2 \times \left(6 + \frac{2}{2} \right) \right\} = 0$$

$$R_B = 10 \text{ kN}$$

(14)

 Page No.
 Date

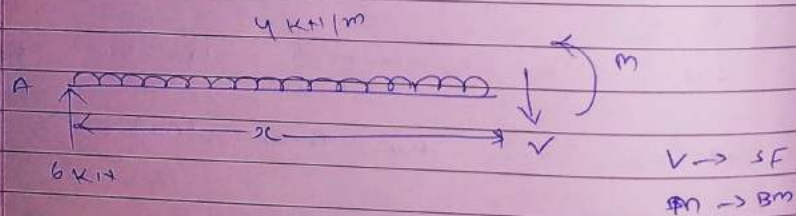
$$\sum F_y = 0$$

$$R_A + R_B - 2 \times (4 \times 2) = 0$$

$$R_A = 16 - 10$$

$$R_A = 6 \text{ kN}$$

→ Take a section b/w (AD) at a distance (x) from (A):-



$$\sum F_y = 0$$

$$6 - V - 4x = 0$$

$$V = 6 - 4x$$

$$\sum M_x = 0$$

$$M - 6 \times (x) + 4 \times (x) \left(\frac{x}{2} \right) = 0$$

(15)

 Page No.
 Date

$$M = 6x - 2x^2$$

$$\text{at } x = 0 :-$$

$$V_A = 6 \text{ kN}$$

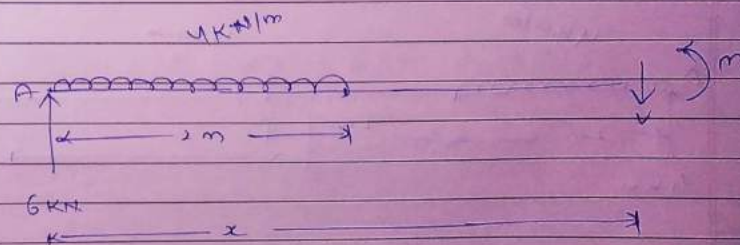
$$M_A = 0$$

$$\text{at } x = 2 \text{ m} :-$$

$$V_D = -2 \text{ kN}$$

$$M_D = 4 \text{ kN-m}$$

→ Take a section (DC) at a distance (x) from (A):-



$$\sum F_y = 0$$

$$6 - 4(x) - V = 0$$

$$V = -2 \text{ kN}$$

$$\sum M_x = 0$$

$$M - 6 \times (x) + 4(x) \left(\frac{x}{2} \right) = 0$$

$$M = 6x - 2x^2$$

(16)

Page No.	
Date	

at $x = 2\text{m}$:-

$$V_D = -2\text{KN}$$

$$M_D = 6(2) - 8(2-1)$$

$$M_D = 4\text{KN-m}$$

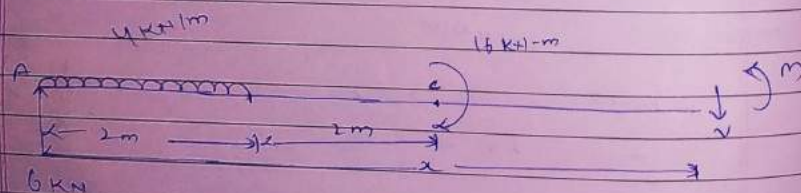
at $x = 4\text{m}$:-

$$V_C = -2\text{KN}$$

$$M_C = 6(4) - 8(4-1)$$

$$M_C = 0$$

⇒ Take a section CE at a distance (x) from A:-



$$\sum F_y = 0$$

$$6 - 4(x) - V = 0$$

$$V = -2\text{KN}$$

$$\sum M_x = 0$$

(17)

Page No.	
Date	

$$M - 6x - 16 + 4(2)(x-1) = 0$$

$$M = 6x - 8(x-1) + 16$$

at $x = 4\text{m}$

$$V_C = -2\text{KN}$$

$$M_C = 6(4) - 8(4-1) + 16$$

$$M_C = 16\text{KN-m}$$

at $x = 6\text{m}$:-

$$V_E = -2\text{KN}$$

$$M_E = 6(6) - 8(6-1) + 16$$

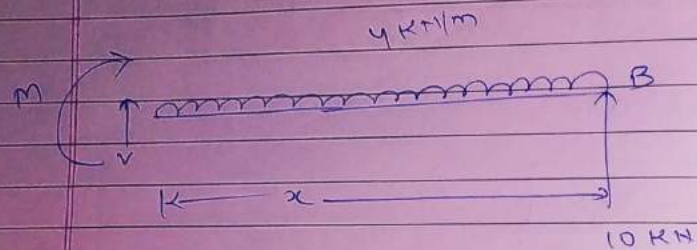
$$M_E = 12\text{KN-m}$$

(18)

Page No.

Date

⇒ Taking section b/w (BE) at a distance (x) from (B):-



$$\sum F_y = 0$$

$$V + 10 - 4x = 0$$

$$V = 4x - 10$$

$$\sum M_x = 0$$

$$-M - 4(x)(x/2) + 10x = 0$$

$$M = -2x^2 + 10x$$

at $x = 0$:-

$$V_B = -10 \text{ kN}$$

$$M_B = 0$$

(19)

Page No.

Date

at $x = 2 \text{ m}$:-

$$V_E = 4(2) - 10$$

$$V_E = -2 \text{ kN}$$

$$M_E = -2(2)^2 + 10 \times (2)$$

$$M_E = 12 \text{ kN-m}$$

Let (a) be the distance from (A) where shear force is zero.

$$0 = 6 - 4a$$

$$a = 1.5$$

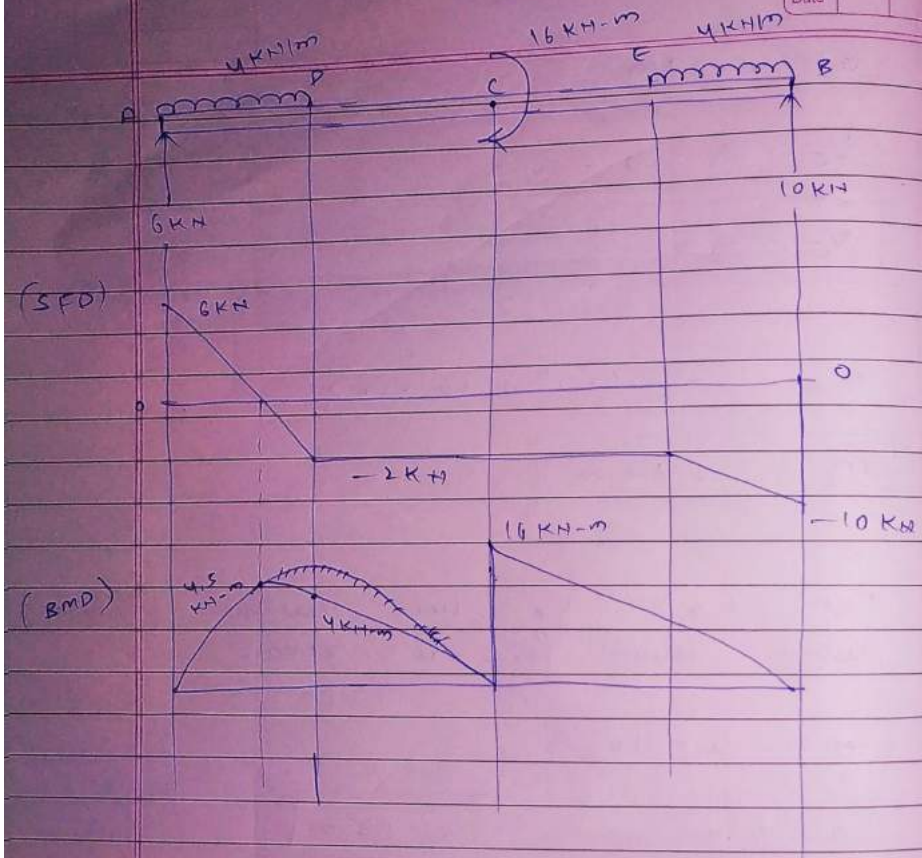
$$a = 1.5 \text{ m}$$

Bm at $x = 1.5$

$$M_a = 6(1.5) - 2(1.5)^2$$

$$M_a = 4.5 \text{ kNm}$$

20

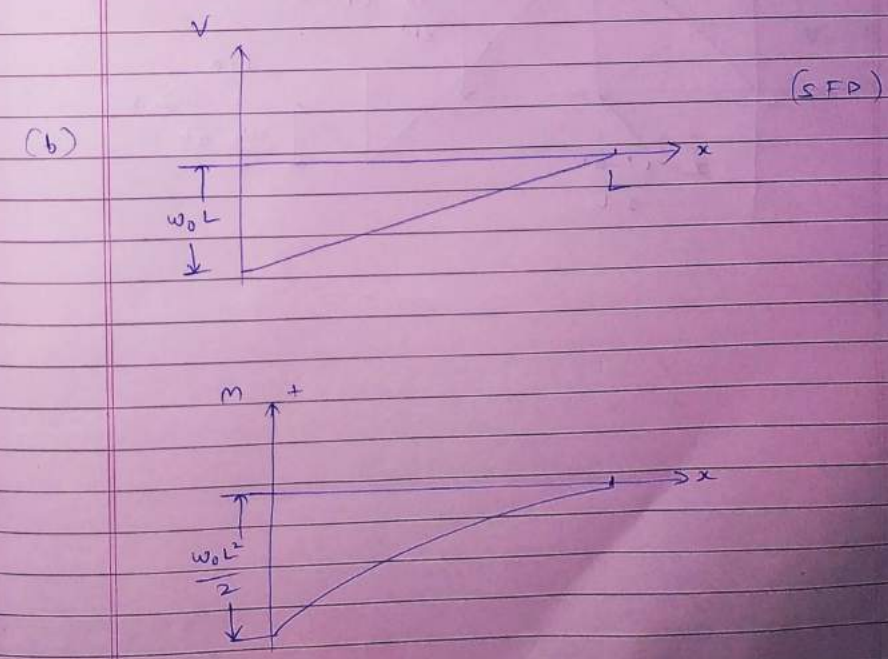
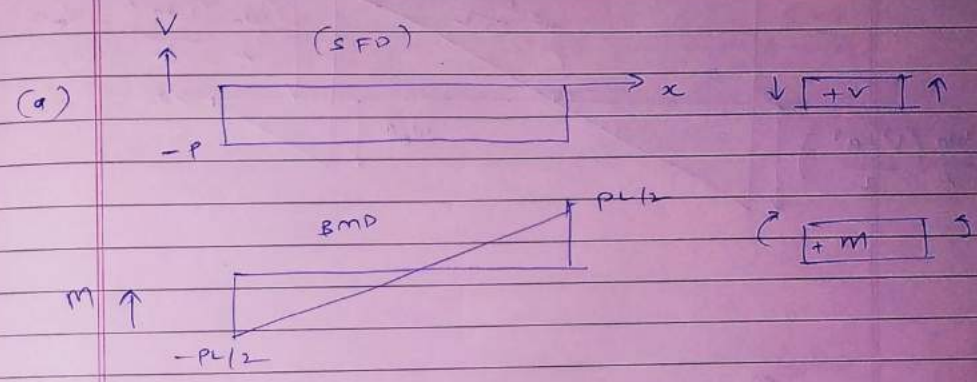


$$V_{max} = 10 \text{ kN}$$

$$M_{max} = 15 \text{ kN-m}$$

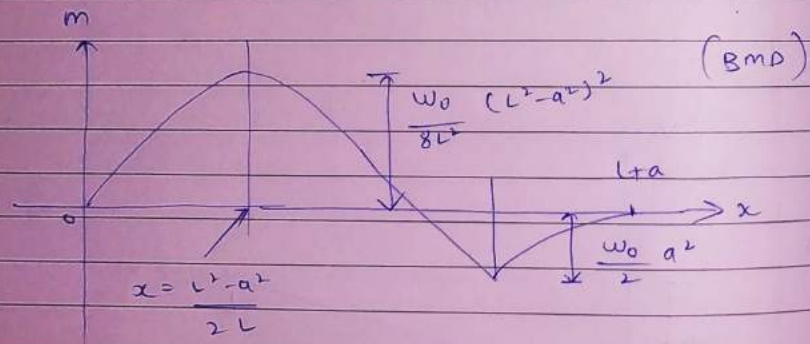
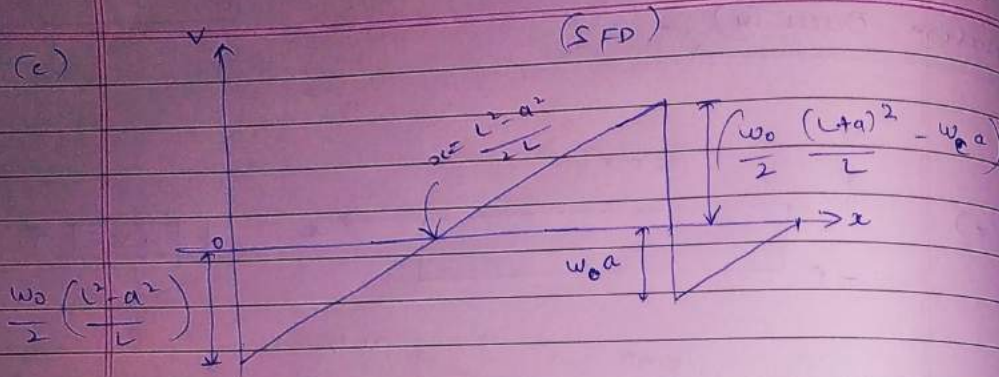
21

Solution Ques (4):-



22

Page No.	
Date	



23

Page No.	
Date	

